MATH329 · OPERATIONS RESEARCH

Midterm 461 Practice

10/10/2024 · 25 marks

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01 Questions

Answer each question in the space below it. The full solution is in a separate file.

Q1 THEORY · 4 M

Answer the following:

a) The formal activities of Operations Research (OR) were initiated in _____ during World War II.

b) An optimization problem is called _____ when the objective function and all the constraints are linear functions of the decision variables.

c) To solve a LPP algebraically, we first put it in the Standard Form. What is this means?

Q2 GRAPHICAL · 6 M

Solve the following problem graphically:

Maximum $z = 10x_1 + 5x_2$

S.to: $x_1 - x_2 \leq 1$

$3x_1 + x_2 \geq 7$

$x_2 \leq 3$

$x_1 \geq 0, x_2 \geq 0$

Q3 STANDARD FORM · 6 M

Given the following problem:

Minimize $Z = 7x_1 + 3x_2$

S.to $2x_1 + x_2 \leq 8$

$x_1 - 3x_2 \geq 12$

$4x_2 = -3$

$x_1 \geq 0$, x_2 unrestricted in sign.

1. What methods can be applied to solve the given linear programming problem?

2. Write the problem in standard form (equation form). "Don't solve the problem"

Q4 STANDARD FORM + SIMPLEX · 9 M

Given the following problem:

Maximize $Z = 8x_1 + 7x_2$

S.to $20x_1 + 16x_2 \leq 62$

$-x_1 + x_2 \leq 1$

$x_1 \leq 3$

$x_1 \geq 0, x_2 \geq 0$

a) Write the problem in standard form (equation form).

b) Solve the problem by the simplex method (stop at the first iteration "Only two tables").

Basic	x_1	x_2	s_1	s_2	s_3	Sol.
Z						
Z						